Upscale

Project definition

B.Sc. in Computer Science, Deep Learning Specialization

Team Members:

1. Moriel Turgeman - 208322305
2. Itay Aknin - 322959412
3. Roi Forer – 322226952

Project Supervisor:

Dr. Moshe Butman

Institution:

The College of Management – Academic Studies
Faculty of Computer Science
Deep Learning Specialization

Git Repository:

https://www.github.com/languax/upscale-dl

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1.1 Background and Motivation

The rapid advancement of display technologies in recent years has significantly increased the demand for high-quality visual content. Modern screens support high resolutions such as Full HD and 4K, exposing visual imperfections that were previously less noticeable. As a result, many legacy and archival video materials—such as old films, documentaries, and historical recordings—appear degraded when viewed on contemporary devices.

These videos often suffer from multiple quality issues, including low spatial resolution, blur, sensor noise, compression artifacts, and temporal inconsistencies. Such degradations originate from limitations in historical recording equipment, aggressive compression standards, aging storage media, and format conversions over time. Traditional video enhancement techniques, such as interpolation-based upscaling, denoising filters, or sharpening operators, provide limited improvement and often introduce additional artifacts or oversmoothing effects.

Recent progress in **deep learning**, particularly in **computer vision**, has enabled data-driven approaches to image and video restoration. Neural networks trained on large datasets have demonstrated the ability to learn complex mappings from degraded visual content to high-quality representations. In the context of **video super-resolution and restoration**, deep learning models can exploit both spatial information within frames and temporal information across consecutive frames, leading to substantial improvements over classical methods.

Despite the existence of commercial solutions and research prototypes, many available systems operate as closed “black-box” tools, offering limited transparency, configurability, or adaptability to specific types of degraded content. Furthermore, research-oriented implementations are often not designed as complete end-to-end systems and require significant engineering effort to apply to real-world video data.

The motivation behind this project is to explore and implement a **deep-learning–based video restoration pipeline** that is both scientifically grounded and practically applicable. By focusing on a well-defined domain of degraded video content and by systematically evaluating neural network architectures and training strategies, this project aims to bridge the gap between theoretical research in video restoration and usable, end-to-end enhancement systems.

1.2 Problem Statement

Many legacy video materials suffer from a combination of degradations that significantly reduce their visual quality when displayed on modern devices. These degradations include low spatial resolution, noise, blur, compression artifacts, and temporal inconsistencies between consecutive frames. The presence of multiple, interacting degradation factors makes classical video enhancement techniques insufficient, as they typically address only isolated aspects of the problem and rely on fixed, hand-crafted rules.

From a learning perspective, the core challenge lies in recovering a high-quality video signal from a degraded version while preserving the original content, structure, and temporal coherence. Unlike single-image restoration, video restoration introduces additional complexity due to motion between frames, occlusions, and the need for temporal consistency to avoid flickering and instability in the output.

Formally, the problem addressed in this project can be defined as follows: given a low-quality input video composed of a sequence of degraded frames, the goal is to reconstruct a corresponding high-quality video sequence that approximates the original, unobserved clean signal. The reconstructed output should exhibit improved sharpness, reduced noise and artifacts, and visually consistent motion across frames, without altering the semantic content of the video.

A major difficulty in addressing this problem is the lack of paired real-world datasets containing both degraded and high-quality versions of the same old videos. As a result, the problem must be approached using a supervised learning framework with synthetically generated degradations that approximate real-world video corruption. The effectiveness of the solution therefore depends not only on the chosen neural network architecture, but also on the realism of the degradation process and the robustness of the training strategy.

In this project, the problem is treated as a **video restoration and super-resolution task**, where the system learns a mapping from low-quality video inputs to high-quality outputs. The challenge is to design, train, and evaluate deep neural networks that can generalize from synthetic training data to real degraded videos, while maintaining both spatial fidelity and temporal consistency.

1.3 Project Objectives

The primary objective of this project is to design, implement, and evaluate a **deep-learning–based system for video restoration and super-resolution**, aimed at improving the visual quality of degraded and low-resolution videos while preserving their original content and motion characteristics.

To achieve this goal, the project defines the following specific objectives:

1. **Formulate the video restoration task as a supervised learning problem**, where low-quality video sequences are mapped to their corresponding high-quality versions using paired training data.
2. **Construct a realistic training dataset** by applying a synthetic degradation process to high-quality videos, simulating common artifacts found in old and legacy video content such as low resolution, noise, blur, and compression artifacts.
3. **Design and evaluate deep neural network architectures** for video restoration and super-resolution, starting from a baseline (naive) model and extending to more advanced architectures, in order to analyze their performance and limitations.
4. **Define and apply appropriate loss functions** that balance pixel-level accuracy, perceptual quality, and temporal consistency across video frames.
5. **Train and fine-tune the proposed models** using the constructed dataset, and study the impact of architectural choices and training strategies on restoration quality.
6. **Evaluate the performance of the trained models** using quantitative metrics (e.g., PSNR, SSIM, perceptual metrics) and qualitative visual analysis, comparing different approaches and configurations.
7. **Develop an end-to-end inference pipeline** that integrates the trained model into a practical system capable of processing full video sequences, including frame extraction, model inference, and video reconstruction.
8. **Analyze the generalization ability of the models** by applying them to real degraded videos and assessing their robustness to degradations not seen explicitly during training.

Through these objectives, the project aims not only to produce visually improved video outputs, but also to provide a systematic investigation of deep learning techniques for video restoration, highlighting the trade-offs between model complexity, performance, and practical applicability.

1.4 Scope, Assumptions and Constraints

This project focuses on the development and evaluation of deep-learning models for video restoration and super-resolution within a clearly defined scope. Establishing explicit assumptions and constraints is essential to ensure feasibility, reproducibility, and meaningful evaluation of the proposed approach.

Scope

The scope of the project includes:

- Enhancing the visual quality of degraded video content through super-resolution and restoration techniques.
- Focusing on videos with **moderate motion and relatively stable scenes**, such as old films, archival footage, and documentary-style recordings.
- Training and evaluating deep learning models using **synthetically degraded data** derived from high-quality video sources.
- Comparing multiple neural network architectures and training strategies to assess their effectiveness in video restoration tasks.
- Implementing an end-to-end inference pipeline capable of processing complete video sequences.

The project does not aim to modify or alter the semantic content of the videos, but solely to improve their visual quality.

Assumptions

The project is based on the following assumptions:

- Synthetic degradations applied to high-quality videos can reasonably approximate real-world degradation patterns found in legacy video content.
- The selected evaluation metrics provide a meaningful indication of perceived visual quality improvements.
- Sufficient computational resources (e.g., GPU access) are available to train and evaluate deep learning models within the project timeline.
- Improvements observed on synthetically degraded data will generalize, at least partially, to real degraded videos.

Constraints

The project operates under several practical constraints:

- Limited training time and computational resources restrict the size and complexity of the models that can be explored.
- The availability of paired real-world low-quality and high-quality video datasets is limited, necessitating reliance on synthetic data generation.
- The project timeline constrains the number of architectures and hyperparameter configurations that can be evaluated.
- The system is designed for offline processing rather than real-time video enhancement.

By explicitly defining these constraints and assumptions, the project maintains a realistic and focused research scope, enabling systematic experimentation and meaningful conclusions.

1.5 Target Domain and Use Cases

The project targets a specific domain of degraded video content in order to ensure focused experimentation and meaningful evaluation. Rather than attempting to address all possible types of video degradation, the project concentrates on a well-defined subset of legacy videos that share common characteristics.

Target Domain

The primary target domain of this project consists of:

- Old cinematic films and archival video footage.
- Videos originally recorded or later converted to low spatial resolutions (e.g., 360p–480p).
- Content affected by noise, blur, and compression artifacts due to historical recording limitations and format conversions.
- Videos with moderate motion and relatively stable scenes, such as dialogue-driven scenes, interviews, documentaries, and historical recordings.

This domain was selected because it represents a common and practical use case for video restoration, while remaining feasible within the constraints of a final-year deep learning project. The relatively moderate motion present in such videos allows for effective exploitation of temporal information without requiring overly complex motion modeling.

Use Cases

The proposed system is designed to support the following use cases:

- **Restoration of old movies and documentaries** for improved viewing on modern high-resolution displays.
- **Enhancement of archival footage** for educational, historical, or cultural preservation purposes.
- **Quality improvement of digitized legacy videos** that suffer from resolution loss and compression artifacts.
- **Experimental evaluation of video restoration models**, enabling systematic comparison of architectures and training strategies.

By clearly defining the target domain and use cases, the project establishes realistic expectations for the system's capabilities and provides a consistent context for dataset construction, model evaluation, and result interpretation.

2. Related Work

2.1 Classical Video Enhancement Methods

Classical video enhancement methods rely on deterministic, hand-crafted algorithms designed to improve specific aspects of visual quality. Common techniques include interpolation-based upscaling methods such as bilinear and bicubic interpolation, spatial filtering for noise reduction, and edge-enhancement or sharpening operators. In some cases, temporal smoothing is applied to reduce frame-to-frame noise fluctuations.

These approaches offer several advantages, including low computational cost, simplicity, and ease of integration into video processing pipelines. As a result, they are widely used in traditional video players and editing software as baseline enhancement tools.

However, classical methods exhibit fundamental limitations when applied to heavily degraded or low-resolution videos. Interpolation-based upscaling cannot recover high-frequency details that are not present in the original signal, often resulting in blurred outputs. Noise reduction filters tend to oversmooth textures and fine structures, while sharpening techniques may amplify noise and compression artifacts. Furthermore, most classical approaches treat frames independently or rely on simplistic temporal assumptions, leading to temporal inconsistencies such as flickering.

Due to their reliance on fixed rules and lack of data-driven learning, classical video enhancement methods are unable to adapt to diverse degradation patterns or complex visual content. These limitations motivate the exploration of learning-based approaches, which can leverage large datasets to model the underlying relationship between degraded and high-quality video content.

2.2 Deep Learning for Image and Video Restoration

Recent advances in deep learning have significantly improved the state of the art in image and video restoration tasks. Convolutional neural networks (CNNs) have demonstrated the ability to learn complex mappings from degraded visual inputs to high-quality outputs by exploiting large amounts of training data. In the context of image super-resolution, learning-based methods outperform classical interpolation techniques by reconstructing fine textures and high-frequency details that are otherwise lost.

Building on image-based approaches, deep learning methods for video restoration extend the learning process to incorporate temporal information across consecutive frames. By leveraging motion cues and temporal correlations, video super-resolution models can improve spatial quality while maintaining temporal consistency. Various architectures have been proposed to address this challenge, including frame-recurrent models, optical-flow–based alignment methods, and spatio-temporal feature aggregation networks.

Representative deep learning models in this domain include convolutional architectures for image super-resolution, generative adversarial networks (GANs) for perceptual quality enhancement, and more recent transformer-based models designed to capture long-range temporal dependencies. While these approaches have achieved impressive results, they often involve trade-offs between reconstruction accuracy, perceptual quality, and computational complexity.

Despite their strong performance, many deep learning–based video restoration methods are primarily developed as research prototypes. Applying these models to real-world video data typically requires careful dataset preparation, degradation modeling, training strategy design, and system-level integration. These challenges highlight the importance of studying not only individual architectures, but also the complete pipeline required to train and deploy video restoration models effectively.

2.3 Positioning of the Proposed Project

The proposed project is positioned at the intersection of academic research in deep learning–based video restoration and practical system development. Rather than introducing a novel state-of-the-art architecture, the focus of this work is on the systematic evaluation and application of existing deep learning models within a well-defined problem setting.

The project emphasizes the construction of a realistic training pipeline, including synthetic degradation modeling and dataset preparation, as well as the comparative analysis of different neural network architectures and training strategies. By starting with a baseline model and progressively exploring more advanced approaches, the project aims to study the trade-offs between model complexity, restoration quality, and computational requirements.

In contrast to commercial “black-box” solutions, the proposed system provides full transparency and control over the learning process, enabling detailed analysis of model behavior and performance. At the same time, unlike purely theoretical research prototypes, the project incorporates an end-to-end inference pipeline capable of processing full video sequences.

Through this balanced approach, the project contributes an applied deep learning perspective on video restoration, focusing on reproducibility, interpretability, and practical applicability rather than solely on achieving maximum benchmark performance.

3. Functional Description and Requirements

3.1 System Overview and End-to-End Flow

The proposed system operates as an end-to-end pipeline for video restoration and super-resolution. Given a low-quality input video, the system processes the video through a sequence of stages that include frame extraction, deep-learning–based enhancement, and video reconstruction. The enhanced video is then returned to the user as a single high-quality output file.

From a functional perspective, the system is designed to support offline video processing, where complete video sequences are processed rather than individual frames in isolation. This design choice enables the use of temporal information across frames and ensures consistency in the restored output.

3.2 AI-Oriented Functional Requirements

From a deep learning perspective, the system must satisfy the following requirements:

- Accept degraded video frames as input and produce enhanced frames with improved spatial resolution and reduced visual artifacts.
- Exploit both spatial and temporal information to improve restoration quality and maintain temporal consistency across frames.
- Support supervised training using paired low-quality and high-quality video data.
- Allow evaluation and comparison of different neural network architectures and training configurations.
- Generalize from synthetically degraded training data to real-world degraded videos.

These requirements define the functional expectations from the learning models independent of their specific architectural implementation.

3.3 System-Level Functional Requirements

At the system level, the solution must:

- Allow users to upload video files for processing.
- Manage video processing tasks and provide feedback on job status.
- Execute model inference using GPU acceleration.
- Reassemble enhanced frames into a final video output.

These system-level requirements serve as supporting infrastructure for the core deep learning components of the project.

4. Architecture

4.1 Overall System Architecture

The proposed system is designed as an end-to-end pipeline for enhancing the visual quality of degraded video content using deep learning. At a high level, the system receives a low-quality video as input and produces an enhanced, high-quality video as output, without modifying the original semantic content.

The overall architecture consists of the following main stages:

1. **Video Input**
   A low-quality video file is provided to the system. The video may suffer from low resolution, noise, blur, and compression artifacts.
2. **Frame Extraction**
   The input video is decomposed into a sequence of individual frames. This step enables frame-based processing while preserving the temporal order of the video.
3. **Deep Learning–Based Enhancement**
   Consecutive frames are processed by a trained deep neural network that performs video restoration and super-resolution. The model enhances spatial details, reduces noise and artifacts, and leverages temporal information across frames to maintain visual consistency.
4. **Frame Reconstruction**
   The enhanced frames produced by the model are collected and reordered according to their original temporal sequence.
5. **Video Output**
   The reconstructed frames are reassembled into a single enhanced video file, which can be viewed or downloaded by the user.

Conceptually, the data flow of the system can be summarized as:

Low-Quality Video → Frame Extraction → Deep Learning Model → Enhanced Frames → Reconstructed Video

This architecture separates the learning-based component from the surrounding video processing stages, allowing the deep learning model to be trained, evaluated, and replaced independently of the system infrastructure. Such a design supports experimentation with different models while maintaining a consistent end-to-end workflow.

4.2 Deep Learning Pipeline

Problem Setup

The learning task is defined as follows:

- **Input:** Short sequences of consecutive video frames extracted from a low-quality video. These frames contain degradations such as reduced resolution, noise, blur, and compression artifacts.
- **Output:** Corresponding high-quality video frames with improved sharpness, reduced noise, and fewer visual artifacts.

Rather than processing each frame independently, the model operates on **small temporal windows** (e.g., 3–7 consecutive frames). This design allows the network to exploit temporal information, such as motion and redundancy between neighboring frames, which is essential for improving restoration quality and avoiding temporal artifacts like flickering.

Supervised Learning Approach

Due to the lack of paired real-world data, the system is trained using synthetically generated training pairs. High-quality video clips serve as ground truth, while their degraded versions are used as inputs. During training, the network learns to reverse the degradation process and reconstruct visually enhanced frames.

This supervised formulation provides:

- Full control over the degradation process.
- Clear training targets.
- A reliable framework for comparing different model architectures and training strategies.

Training vs. Inference Pipeline

The pipeline is conceptually divided into two modes:

- **Training Mode:**
  Video clips are sampled from the dataset, degraded synthetically, and fed into the model along with their original high-quality versions. Model parameters are optimized using a predefined loss function.
- **Inference Mode:**
  The trained model is applied to unseen low-quality videos. Frames are processed sequentially in overlapping windows to ensure temporal consistency across the entire video.

4.3 Dataset and Data Preparation

A central challenge in video restoration tasks is the lack of real-world datasets containing paired low-quality and high-quality versions of the same video content. To address this limitation, the project relies on a **synthetic data generation approach**, which enables supervised training under controlled conditions.

High-Quality Data Sources

The dataset construction process begins with the collection of high-quality (HQ) video content. These videos serve as the ground-truth reference during training. The selected videos are characterized by:

- High spatial resolution (e.g., Full HD or higher).
- Relatively stable scenes with moderate motion.
- Minimal compression artifacts and noise.

Such content can be obtained from publicly available video datasets, open-source cinematic footage, or modern high-quality recordings. Using clean, high-resolution videos ensures that the model learns to reconstruct visually plausible and detailed outputs.

Synthetic Degradation Process

To generate corresponding low-quality (LQ) inputs, each high-quality video is passed through a synthetic degradation pipeline that simulates common artifacts found in old and legacy video content. The degradation process includes combinations of the following operations:

- **Downscaling:** Reducing spatial resolution to simulate low-resolution recordings (e.g., from 1080p to 360p or 480p).
- **Noise Injection:** Adding Gaussian or similar noise to emulate sensor noise and analog artifacts.
- **Blurring:** Applying mild blur to simulate focus issues and motion blur.
- **Compression Artifacts:** Encoding videos using aggressive compression settings to introduce block artifacts and ringing effects.

By combining these degradations, the generated low-quality videos approximate the visual characteristics of real degraded footage while maintaining a known high-quality reference.

Training Data Structure

For training purposes, videos are divided into short clips consisting of consecutive frames. Each training sample includes:

- A sequence of degraded frames (LQ) as input.
- The corresponding sequence of original high-quality frames (HQ) as the target.

The dataset is split into **training**, **validation**, and **test** subsets to enable proper evaluation and prevent overfitting. The validation set is used for model selection and hyperparameter tuning, while the test set is reserved for final performance evaluation.

Data Preparation Pipeline

Before being fed into the neural network, video frames undergo basic preprocessing steps such as normalization and formatting into tensor representations. Data loading is designed to preserve the temporal order of frames within each clip, which is critical for learning temporal relationships.

This dataset preparation strategy enables systematic experimentation with different models and training configurations, while maintaining full control over the learning process and evaluation conditions.

4.4 Model Architecture (Baseline)

The baseline model architecture is designed to provide a simple, interpretable, and effective starting point for the video restoration task. Its purpose is to establish a reference performance level against which more advanced models and design choices can be evaluated later in the project.

Design Principles

The baseline architecture is guided by the following principles:

- **Simplicity:** The model should be easy to understand and implement, without relying on complex or specialized components.
- **Feasibility:** The architecture must be trainable within the available computational resources and project timeline.
- **Temporal Awareness:** The model should utilize limited temporal information to improve restoration quality compared to single-frame processing.

Model Structure

The baseline model is based on a convolutional neural network (CNN) that operates on short sequences of consecutive video frames. A fixed number of neighboring frames is provided as input, forming a small temporal window centered around a target frame.

The input frames are concatenated along the channel dimension and processed through a series of convolutional layers that extract and refine spatio-temporal features. The network outputs a single enhanced frame corresponding to the central frame of the input sequence.

This design allows the model to:

- Leverage information from neighboring frames to recover missing details.
- Reduce noise and artifacts by exploiting temporal redundancy.
- Maintain a relatively low computational complexity.

Output Resolution

The baseline model is designed to perform both restoration and spatial upscaling. Depending on the chosen configuration, the model may output frames at the original input resolution or at a higher resolution (e.g., 2× upscaling). This flexibility enables experimentation with different super-resolution settings.

Role of the Baseline Model

The baseline architecture serves as a foundation for experimentation throughout the project. Its performance provides a benchmark against which alternative architectures, extended temporal windows, or modified training strategies can be compared.

By starting with a baseline model, the project ensures a systematic and controlled approach to model development, allowing performance improvements to be attributed to specific architectural or methodological changes.

4.5 Loss Function

Training a deep learning model for video restoration requires a loss function that accurately reflects the desired visual improvements in the output. The loss function guides the optimization process by quantifying the difference between the restored frames produced by the model and the corresponding high-quality ground-truth frames.

Primary Reconstruction Loss

The primary loss used in this project is a **pixel-wise reconstruction loss**, such as **L1 loss** (mean absolute error). This loss measures the absolute difference between the predicted frame and the ground-truth frame on a per-pixel basis.

L1 loss is chosen due to its stability and robustness when training restoration models. Compared to L2 loss, it is less sensitive to outliers and tends to preserve sharper edges, which is important for visual quality in super-resolution tasks.

Perceptual Considerations

While pixel-wise losses encourage accurate reconstruction, they may not fully capture perceptual quality as perceived by human observers. To address this, the project considers the use of an additional **perceptual loss** component, computed on high-level feature representations extracted from a pretrained convolutional network.

This perceptual loss encourages the restored output to match the ground-truth not only at the pixel level, but also in terms of visual structure and texture. Its inclusion allows experimentation with trade-offs between numerical accuracy and perceptual quality.

Loss Function Composition

The overall training objective is defined as a weighted combination of loss terms, for example:

- A dominant reconstruction loss (L1)
- An optional perceptual loss component

The relative weights of these components can be adjusted during experimentation to study their effect on restoration quality and training behavior.

Temporal Consistency Considerations

Although the baseline model focuses primarily on spatial reconstruction, temporal consistency is implicitly encouraged by training on consecutive frame sequences. More explicit temporal consistency losses may be explored in later stages of the project as part of architectural or methodological extensions.

Role of the Loss Function in the Project

The loss function plays a central role in determining the behavior and output quality of the model. By starting with a simple and well-understood loss formulation and gradually exploring extensions, the project maintains a controlled and interpretable training process.

4.6 Inference and Output Generation

After the model has been trained, it is applied in inference mode to enhance real low-quality videos. The inference process is designed to operate on complete video sequences while preserving temporal order and visual consistency.

Video Processing at Inference Time

During inference, an input video is first decomposed into a sequence of frames, similar to the training process. Frames are processed sequentially using the trained baseline model, which operates on short temporal windows of consecutive frames. For each window, the model produces an enhanced version of the central frame.

To ensure full coverage of the video, overlapping windows are used so that each frame is restored using information from its neighboring frames. This approach helps maintain temporal consistency across the entire video and reduces artifacts such as flickering.

Handling Video Boundaries

At the beginning and end of the video, where fewer neighboring frames are available, simple strategies such as frame replication or reduced window sizes are applied. These techniques allow consistent processing without introducing special-case behavior in the model itself.

Frame Reconstruction and Video Assembly

Once all frames have been processed and enhanced, the restored frames are reassembled in their original temporal order. The final output is generated as a single video file using standard video encoding techniques.

The output video preserves the original frame rate and temporal structure of the input, while exhibiting improved spatial resolution and reduced visual degradations.

Practical Considerations

The inference pipeline is designed for offline processing, allowing the system to handle videos of varying lengths without real-time constraints. GPU acceleration is used during model inference to ensure reasonable processing times.

By clearly separating the inference pipeline from the training process, the system allows trained models to be reused and evaluated on different types of degraded video content without modification.

5. Work Plan

This section outlines the planned stages of the project, from initial setup to final evaluation and submission. The work plan is structured to support an iterative development process, with an emphasis on deep learning experimentation, evaluation, and system integration.

5.1 Project Milestones

The project is divided into the following main milestones:

M1 – Project Setup and Dataset Preparation
This stage includes finalizing the project scope, selecting high-quality video sources, and implementing the synthetic degradation pipeline. The dataset will be organized into training, validation, and test subsets, enabling supervised learning and systematic evaluation.

M2 – Baseline Model Implementation and Training
During this milestone, the baseline deep learning model is implemented and trained using the prepared dataset. Initial experiments are conducted to verify correct learning behavior and establish a reference performance level.

M3 – Model Evaluation and Architectural Experiments
In this phase, the baseline model is evaluated using quantitative metrics and visual inspection. Based on the results, alternative architectural variations and training configurations are explored and compared against the baseline.

M4 – Inference Pipeline and System Integration
This milestone focuses on integrating the trained model into an end-to-end inference pipeline capable of processing full video sequences. Frame extraction, model inference, and video reconstruction are combined into a cohesive workflow.

M5 – Testing, Analysis, and Refinement
The complete system is tested on unseen degraded videos, including real legacy footage. Results are analyzed, limitations are identified, and refinements are applied where feasible.

M6 – Documentation and Final Presentation
The final stage includes completing project documentation, preparing evaluation summaries, and assembling materials for the final submission and presentation.

5.2 Team Responsibilities

Although all team members participate in the deep learning aspects of the project, responsibilities are divided to ensure efficient progress and clear ownership of tasks.

- **Model Development and Experiments:**
  Team members collaboratively design, train, and evaluate deep learning models. Different architectures and training strategies are explored in parallel and compared systematically.
- **Dataset and Degradation Pipeline:**
  Responsibility for collecting high-quality videos, implementing the synthetic degradation process, and managing dataset splits is shared among team members involved in experimentation.
- **System Integration and Inference Pipeline:**
  The implementation of the end-to-end inference pipeline, including frame processing and video reconstruction, is handled collaboratively, ensuring compatibility with the trained models.
- **Evaluation and Analysis:**
  Quantitative evaluation, visual analysis, and comparison of experimental results are conducted jointly, with findings documented and discussed across the team.

This collaborative structure ensures that all team members gain hands-on experience with deep learning while contributing to different aspects of the overall system.

6. Client Side

The client side of the system provides a simple and intuitive interface that allows users to interact with the video restoration pipeline. Its primary role is to enable video submission, track processing progress, and retrieve the enhanced output, while abstracting away the underlying deep learning and system complexity.

6.1 Usage Illustration

The typical user interaction flow is as follows:

1. The user uploads a low-quality video file through the web interface.
2. The system confirms successful upload and initiates the processing task.
3. The user is presented with a job status indicator showing the progress of the video enhancement process.
4. Once processing is complete, the user is able to preview and download the enhanced video.

This flow is designed to be linear and minimal, requiring no technical knowledge of deep learning or video processing from the user.

6.2 UI Design Considerations

The client interface emphasizes clarity and ease of use. Key design considerations include:

- A clean upload interface supporting common video formats.
- Clear feedback during processing, including status updates and completion notifications.
- Simple access to the final enhanced video without additional configuration steps.

The client side does not expose internal model parameters or training options, as its purpose is to demonstrate the practical application of the trained deep learning model rather than to provide advanced configuration capabilities.

7. Server Side

The server side of the system is responsible for orchestrating the video enhancement process and supporting the execution of the deep learning model. It acts as an intermediary between the client interface and the core inference pipeline, handling video uploads, task management, and result delivery.

The server is designed to support offline processing of video content and to manage computational resources efficiently during model inference.

7.1 Backend System Overview

The backend provides a centralized control layer for the system. Its main responsibilities include:

- Receiving video files uploaded by users.
- Managing processing jobs and their execution order.
- Invoking the deep learning inference pipeline.
- Storing intermediate and final results.
- Exposing APIs for status tracking and result retrieval.

The backend is implemented as a modular service, allowing the deep learning components to be updated or replaced without affecting the client interface.

7.2 Inference and Processing Pipeline

Once a video is uploaded, the backend initiates the inference pipeline. The pipeline consists of the following stages:

1. **Video Validation and Storage**
   The uploaded video is validated and stored temporarily on the server.
2. **Frame Extraction**
   The video is decomposed into individual frames while preserving frame order and frame rate.
3. **Model Inference**
   Frames are grouped into short temporal windows and passed to the trained deep learning model for enhancement. GPU acceleration is used during this stage to reduce processing time.
4. **Frame Collection and Reconstruction**
   Enhanced frames are collected and reordered to match the original temporal sequence.

5. **Output Video Generation**
   The restored frames are encoded into a final enhanced video file.

This pipeline allows the server to process videos of varying lengths in a consistent and scalable manner.

7.3 Task Scheduling and Resource Management

Video enhancement is a computationally intensive task, particularly during model inference. To handle this, the backend manages processing jobs using a simple task queue mechanism.

Key considerations include:

- Ensuring that only a limited number of jobs are processed concurrently to avoid GPU overload.
- Tracking the state of each job (e.g., pending, processing, completed, failed).
- Allowing users to query the status of their submitted jobs.

This approach ensures stable system behavior even when multiple videos are submitted for processing.

7.4 API Specification

The backend exposes a small set of APIs that enable communication between the client and server components.

7.4.1 Video Upload API

- **Endpoint:** POST /upload
- **Description:** Accepts a video file from the client and creates a new processing job.
- **Response:** A unique job identifier used for tracking.

7.4.2 Job Status and Progress API

- **Endpoint:** GET /status/{jobId}
- **Description:** Returns the current status of the processing job (e.g., queued, processing, completed).

7.4.3 Result Retrieval API

- **Endpoint:** GET /result/{jobId}
- **Description:** Provides access to the enhanced video once processing is complete.